

Problem 2

Part 2a) i)

Let $n \in \mathbb{N}$ be fixed. For any $\omega \in \Omega = [0, 1)$, we have that $0 \leq n\omega < n$. Then $\lfloor n\omega \rfloor \in \{0, 1, \dots, n-1\}$. Thus, the target set is:

$$X_n(\omega) = \frac{\lfloor n\omega \rfloor}{n} \in \left\{0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}\right\} := E_n.$$

Therefore, the target measurable space for X_n is $(E_n, \mathcal{E}_n)$, where $\mathcal{E}_n = \mathcal{P}(E_n)$.

To show that $X_n : (\Omega, \mathcal{F}) \rightarrow (E_n, \mathcal{E}_n)$ is a random variable, we must verify that it is $\mathcal{F}$ - $\mathcal{E}_n$ -measurable (Definition 3.1.2), i.e., $X_n^{-1}(B) \in \mathcal{F}, \forall B \in \mathcal{P}(E_n)$. For any singleton event $\left\{\frac{k}{n}\right\} \in E_n$ (where $k \in \{0, \dots, n-1\}$), the preimage evaluates to

$$\begin{aligned} X_n^{-1}\left(\left\{\frac{k}{n}\right\}\right) &= \left\{\omega \in [0, 1) \mid \frac{\lfloor n\omega \rfloor}{n} = \frac{k}{n}\right\} \\ &= \{\omega \in [0, 1) \mid \lfloor n\omega \rfloor = k\} \\ &= \{\omega \in [0, 1) \mid k \leq n\omega < k+1\} \\ &= \left[\frac{k}{n}, \frac{k+1}{n}\right). \end{aligned}$$

For $k \in \{0, \dots, n-1\}$, $\left[\frac{k}{n}, \frac{k+1}{n}\right) \in \mathcal{B}([0, 1)) = \mathcal{F}$.

Now let $B \in \mathcal{P}(E_n)$ be an arbitrary event in our target space. Since B is a subset of the finite set E_n , it can be written as a finite union of its singletons:

$$B = \bigcup_{x \in B} \{x\} = \bigcup_{k \in K} \left\{\frac{k}{n}\right\},$$

where $K \subset \{0, \dots, n-1\}$. Applying the preimage operator yields:

$$X_n^{-1}(B) = X_n^{-1}\left(\bigcup_{k \in K} \left\{\frac{k}{n}\right\}\right) = \bigcup_{k \in K} X_n^{-1}\left(\left\{\frac{k}{n}\right\}\right).$$

Since B is finite (and therefore countable) and each component preimage $X_n^{-1}\left(\left\{\frac{k}{n}\right\}\right) = \left[\frac{k}{n}, \frac{k+1}{n}\right) \in \mathcal{F}$, then $X_n^{-1}(B) = \bigcup_{k \in K} X_n^{-1}\left(\left\{\frac{k}{n}\right\}\right) \in \mathcal{F}$ (closure of the σ -algebra $\mathcal{F}$ under countable unions). Thus, X_n is a discrete random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. □

Part 2a) ii)

From part i), we established that $X_n(\omega) \in E_n = \{0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}\}$ and that $X_n^{-1}\left(\left\{\frac{k}{n}\right\}\right) = \left[\frac{k}{n}, \frac{k+1}{n}\right), \left\{\frac{k}{n}\right\} \in E_n$ for $k \in \{0, 1, \dots, n-1\}$. The probability distribution of X_n is the pushforward measure $\mathbb{P}_{X_n} = \mathbb{P} \circ X_n^{-1}$ on the measurable space $(E_n, \mathcal{P}(E_n))$ (Definition 3.1.5). For any $k \in \{0, 1, \dots, n-1\}$, we have:

$$\begin{aligned} \mathbb{P}_{X_n}\left(\left\{\frac{k}{n}\right\}\right) &= \mathbb{P}\left(X_n^{-1}\left(\left\{\frac{k}{n}\right\}\right)\right) \\ &= \lambda_{[0,1)}\left(\left[\frac{k}{n}, \frac{k+1}{n}\right)\right) \\ &= \frac{k+1}{n} - \frac{k}{n} \\ &= \frac{1}{n}. \end{aligned}$$

Since each of the n distinct points in E_n is assigned an equal probability of $\frac{1}{n}$, the random variable X_n follows a discrete uniform distribution (Definition 1.2.9) on E_n . Formally:

$$\mathbb{P}_{X_n}(B) = \frac{1}{n} \sum_{k=0}^{n-1} \delta_{\frac{k}{n}}(B), \quad \forall B \in \mathcal{P}(E_n).$$

□

Part 2b)

We prove this statement using measure-theoretic induction.

Step 1: Indicator Functions. Let $A \in \sigma(X)$ and $Y = 1_A$. Then Y is $\sigma(X)$ - $\mathcal{B}(\mathbb{R})$ -measurable. We have that $A \in \sigma(X) = \{X^{-1}(B) \mid B \in \mathcal{E}\}$. Therefore, $\exists B \in \mathcal{E} : A = X^{-1}(B)$. For any $\omega \in \Omega$, we have:

$$Y(\omega) = 1_A(\omega) = 1_{X^{-1}(B)}(\omega) = 1_B(X(\omega))$$

Let $h := 1_B$, such that $Y(\omega) = h(X(\omega))$ for all $\omega \in \Omega$. Since $B \in \mathcal{E}$, h is $\mathcal{E}$ - $\mathcal{B}(\mathbb{R})$ -measurable.

Step 2: Non-negative Simple Functions. Let Y be a non-negative simple $\sigma(X)$ -measurable function. Let the following be the canonical representation of Y :

$$Y = \sum_{i=1}^n c_i 1_{A_i}$$

where $c_i \geq 0$ and $A_i \in \sigma(X)$. From Step 1, $\forall i \in \{1, \dots, n\} \exists B_i \in \mathcal{E} : A_i = X^{-1}(B_i)$. We define:

$$h := \sum_{i=1}^n c_i 1_{B_i}$$

such that for all $\omega \in \Omega$:

$$Y(\omega) = \sum_{i=1}^n c_i 1_{A_i}(\omega) = \sum_{i=1}^n c_i 1_{X^{-1}(B_i)}(\omega) = \sum_{i=1}^n c_i 1_{B_i}(X(\omega)) = h(X(\omega)).$$

Since $B_i \in \mathcal{E}$ for all $i = 1, \dots, n$, h is a non-negative simple function on $(E, \mathcal{E})$ and, therefore, $\mathcal{E}$ - $\mathcal{B}(\mathbb{R})$ -measurable.

Step 3: Non-negative Measurable Functions.

Let Y be a nonnegative $\sigma(X)$ - $\mathcal{B}(\mathbb{R})$ -measurable function. Then, there exists a sequence of non-negative simple $\sigma(X)$ - $\mathcal{B}(\mathbb{R})$ -measurable functions $(Y_n)_{n \in \mathbb{N}}$ such that $0 \leq Y_n \uparrow Y$. By Step 2, for each $n \in \mathbb{N}$, there exists an $\mathcal{E}$ - $\mathcal{B}(\mathbb{R})$ -measurable simple function h_n such that $Y_n(\omega) = h_n(X(\omega))$. We define $h : E \rightarrow \mathbb{R}$ as:

$$h(x) := \limsup_{n \rightarrow \infty} h_n(x)$$

For all $\omega \in \Omega$, the sequence $(h_n(X(\omega)))_{n \in \mathbb{N}}$ is non-decreasing since $h_n(X(\omega)) = Y_n(\omega) \uparrow Y$, and therefore $\lim_{n \rightarrow \infty} h_n(X(\omega)) = \limsup_{n \rightarrow \infty} h_n(X(\omega))$. Putting everything together:

$$Y(\omega) = \lim_{n \rightarrow \infty} Y_n(\omega) = \lim_{n \rightarrow \infty} h_n(X(\omega)) = \limsup_{n \rightarrow \infty} h_n(X(\omega)) = h(X(\omega)), \quad \forall \omega \in \Omega.$$

Because the limit superior of measurable functions is measurable (Lemma 3.1.9), h is $\mathcal{E}$ - $\mathcal{B}(\mathbb{R})$ -measurable.

Step 4: General Measurable Functions.

Let Y be an arbitrary $\sigma(X)$ - $\mathcal{B}(\mathbb{R})$ -measurable function. We decompose Y into its positive and negative parts, $Y = Y^+ - Y^-$, where $Y^+, Y^- \geq 0$ and both are $\sigma(X)$ - $\mathcal{B}(\mathbb{R})$ -measurable. By Step 3, there exist $\mathcal{E}$ - $\mathcal{B}(\mathbb{R})$ -measurable functions h^+ and h^- such that $Y^+(\omega) = h^+(X(\omega))$ and $Y^-(\omega) = h^-(X(\omega))$. We define:

$$h(x) := \begin{cases} h^+(x) - h^-(x) & \text{if } h^+(x) < \infty \text{ or } h^-(x) < \infty \\ 0 & \text{otherwise} \end{cases}$$

such that for all $\omega \in \Omega$:

$$Y(\omega) = Y^+(\omega) - Y^-(\omega) = h^+(X(\omega)) - h^-(X(\omega)) = h(X(\omega))$$

Since h is the difference of two $\mathcal{E}$ - $\mathcal{B}(\mathbb{R})$ -measurable functions, h is also $\mathcal{E}$ - $\mathcal{B}(\mathbb{R})$ -measurable. □